

## ARive

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# ARive: Assisting Drivers with In-Car Augmented Reality for Risk Zone Detection

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Human factors such as fatigue and distraction often impair drivers' ability to gauge traffic dynamics, leading to collisions, especially at unsignalized intersections. Augmented reality (AR) technology, particularly through advanced 3D projections and wearable head-mounted displays (HMDs), offers a promising enhancement by integrating comprehensive environmental awareness directly into the driver's field of view. This paper presents "ARive," an innovative AR driver-assistance system designed to improve road safety by projecting dynamic risk zones beneath other traffic participants, thus providing real-time kinematic information to promote safer driving distances and informed decision-making. The research involved developing two distinct AR designs and testing them using a fixed-base driving simulator with integrated real-time data communication. A user study with 17 participants revealed that while AR projections significantly improve distance maintenance, particularly in abrupt braking scenarios, they do not markedly affect brake response times or enhance safety during critical events. These findings suggest the need for further optimization of AR design elements to maximize effectiveness, highlighting the potential of AR in enhancing driver awareness and safety.

CCS Concepts: • **Human-centered computing** → **Mixed / augmented reality**.

Additional Key Words and Phrases: driving assistance system, augmented reality, driving simulator, interaction design

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## 1 Introduction

Human drivers may not always be proficient in accurately gauging the kinematics of surrounding traffic participants due to various human factors, for example fatigue, distraction, and overconfidence. This deficiency is one of the leading causes of collisions related to left turn [35, 43, 48]. For example, when two drivers approach an unsignalized intersection simultaneously, traffic rules apply in which one of the drivers has the right of way through the intersection. The driver who does not have the right of way must decide whether the gap is large enough to enter the intersection without causing disruption and forcing the other vehicle to slow down abruptly. Another challenge arises when a driver is following a lead vehicle that suddenly decelerates. In this case, the

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driver has to gauge the intensity of the deceleration by the rate of the lead vehicle appearing larger, as the distance becomes smaller.

Augmented reality (AR) technology, which overlays digital information onto the real world, offers substantial potential for delivering warnings and suggestions about hazardous objects or areas. Recently, the industry has been actively exploring AR's application as an advanced driver-assistance system (ADAS). This implementation could provide drivers with additional information about their surrounding traffic environment, resulting in a better overall driving experience and improved road safety [37].

Contemporary applications of in-car AR-HUD (augmented reality heads-up display) have the capability to provide the driver with relevant 2D information, for example, traffic signs, speed limit, current speed, and navigation. However, these windscreen displays suffer from a limited field of view (FOV) and may lead to disorientation because drivers must frequently shift their gaze from a closer objects (2D information) to the more distant driving environment [28]. To address these limitations, more sophisticated AR applications utilizes 3D AR projections to present images on the entire windscreen and integrate them seamlessly into the surrounding road environment [20, 74]. Nevertheless, it's important to note that these systems are still in a conceptual phase due to the remaining technological limitations [22]. In recent years, there have been notable advancements in the development of wearable AR devices, providing an alternative solution. AR glasses or head-mounted displays (HMD) have the ability to monitor the user's gaze and track their line of sight, leading to a considerably expanded FOV compared to conventional HUD. While a few car manufacturers have explored the integration of AR HMDs for in-vehicle use, these projects have largely remained in the conceptual phase. Furthermore, the majority of the driving simulator studies have simulated AR by overlaying graphics on the monitor or by using virtual reality. Consequently, there is a lack of empirical studies evaluating the practical use of AR HMDs for in-vehicle use.

To further investigate the impact of AR head-mounted displays on safety-related applications, we designed two distinct interfaces specifically tailored for augmented reality (AR) glasses; these interfaces enhance drivers' perception of other traffic participants' kinematics by superimposing a dynamic red area (a risk zone) beneath them through a holographic overlay. In order to verify our concept, we have implemented the interfaces on real augmented reality glasses. Moreover, a fixed-base driving simulator has been developed that enables real-time data communication with the AR device, and multiple driving scenarios were implemented. Then a user study involving 17 participants showed participants were conducted to compare among two designs and non-AR assistance. The quantitative driving data shows that participants behaved safer with AR visualization comparing with non-AR condition. The questionnaires and interviews results show that user's have a positive attitude towards using AR glass for safety applications in vehicle, and prefer the visualizations which not only highlight the risky objects but also project the potential risky area on the road.

Over all, the main contributions of our research laying on the following perspectives:

- We conducted the first user feedback study on a driving assistance application using a real AR headset. Previous research on AR for driving assistance primarily focused on AR-HUDs and windshield displays. Although some studies explored AR-HMDs as an alternative, they did not project content directly onto the surrounding environment. In contrast, our implementation with AR glasses allowed participants to experience high-fidelity AR content projected onto their surroundings, closely resembling a real-world product. This approach offers more robust insights into user acceptance of AR-HMDs for in-vehicle HMI applications, including user experience and mental workload.
- We compared two different designs specifically tailored for real AR-HMD devices, representing distinct strategies for visualizing predictive risks. These designs were tested in various risky scenarios within a driving simulator with participant involvement. Analysis of driving behavior and subjective feedback offers valuable insights for researchers and practitioners to further explore the design space of AR-HMDs in enhancing driving safety.

## 2 Related Work

In this chapter, we first explore how AR is employed in vehicles, focusing on recent developments in head-mounted AR devices. We then review AR interface design for driving assistance, and finally examine prior research on AR's impact on driver distraction.

### 2.1 Platforms for In-Vehicle Augmented Reality

**2.1.1 AR-HUD and Windshield Display.** AR is becoming more standardized in the automotive industry due to its potential to enhance user experience and driver safety. A number of studies [6, 13, 22, 62, 70–72] have explored the technology in various automotive applications, for which the most popular is the AR-HUD to provide vehicle information, navigational aid, and road information.

In a study by Charissis et al. [11], an AR HUD has been developed that projected incoming information onto the driver's point of view (POV). The researchers investigated it in contrast to a traditional HDD, using a gesture recognition interface. In a fixed-base driving simulator study (n=30), the researchers found a 64% increase in collision avoidance through the maintenance of eye gaze on the road. Medenica et al. [44] investigated the upcoming AR personal navigation device (PND) in comparison to the traditional street view PND and map based PND. Through a fixed-base driving simulator study, it was found that the AR PND increased the eye gaze time on the road ahead, improving visual attention and driving performance. Furthermore, the subjective response from the participants indicated a largely positive attitude. Another study on AR PND by Bark et al. (2014) [2] conducted a fixed-base driving simulator study on a developed 3D volumetric HUD to aid the driver in navigating. The evaluation by 15 users revealed an enhanced turn recognition time in comparison to traditional PND. Supporting the previous study, they found that the use of an AR PND helped the driver fixating on the driving environment more. Additionally, the researchers argued in a second study, comparing 2D-HUD and 3D-HUD, a necessity for the implementation of a dynamically adjustable focal plane (3D-HUD) for proper registration of the AR cues.

The leading manufacturers of HUD for the automotive industry, eg., Bosch, Continental, and Panasonic have successfully penetrated the market by developing bright tablet-sized windscreen projectors [27]. However, These windscreen displays have a limited field of view (FOV) and projection size. Emerging developments in AR by companies such as WayRay [74] and EyeLights [20] may overcome these limitations by allowing full AR windscreen projection into future vehicles for not only user experience, but also road safety. Nevertheless, a full AR windscreen display is still at a conceptual stage for which the remaining technological challenges prevent the implementation of this technology in near future vehicles [22].

**2.1.2 AR-HMD.** In recent years, there has been significant advancements in the development of wearable AR-HMD devices, which offer an alternative solution. AR glasses or head-mounted displays have the capability to track the user's gaze and follow their line of sight, resulting in a significantly larger field of view (FOV). In the automotive domain, the technology holds the potential to project relevant information onto objects in different lanes, sidewalks, or even traffic signs.

Several car manufacturers, for example Volkswagen and Mercedes-Benz, have made efforts to link AR glasses to in-vehicle systems to offer functionalities similar to a full AR windscreen display. BMW has built extensive experience with AR/MR devices, showcased through projects like the MINI Augmented Vision<sup>1</sup>, collaborations with Meta Reality Labs<sup>2</sup> and BMW Motorrad's ConnectedRide Smartglasses<sup>3</sup>. The applications ranges from gaming, navigation to parking. In the academic field, an initial study using AR-HMDs for in-vehicle HMI was

<sup>1</sup><https://www.press.bmwgroup.com/global/article/detail/T0213150EN/mini-augmented-vision-a-revolutionary-display-concept-enhances-safety-and-convenience>

<sup>2</sup><https://www.press.bmwgroup.com/deutschland/article/detail/T0417479DE/bmw-group-und-meta-reality-labs-stellen-gemeinsame-forschung-zur-vernetzung-von-extended-reality-geraeten-mit-digitaalem-fahrzeug-oekosystem-vor?language=de>

<sup>3</sup><https://www.press.bmwgroup.com/global/article/detail/T0422306EN/bmw-motorrad-presents-connectedride-smartglasses>

conducted by Kun et al. [33], who analyzed driver behavior during video calls with the aid of a HoloLens 1 and a stationary driving simulator. More recently, He et al. [23] employed lightweight AR glasses to explore navigation applications. However, if the visualized content was not properly registered at the correct location, this is by definition not a real AR application.

In summary, while the swift advancement of AR wearables garners significant interest from researchers and industry professionals in the automotive domain, the industry's exploration remains largely at the conceptual level, and academic research is relatively scarce.

## 2.2 AR Interface Design for Driving Assistance

Augmented Reality (AR) technology has been extensively adopted to enhance driving safety by projecting information directly onto the surroundings. Although there are various types of display devices, interface designs generally fall into two main categories: either they directly highlight hazardous objects or they demarcate dangerous areas.

**2.2.1 Visualizing dangerous objects.** Directly highlighting the dangerous object can quickly indicate driver the source of the danger for further measurement. For example, an AR interface proposed by FCWS by Phan et al. [52] was focused on marking the pedestrians which may collide with ego vehicle. The system has been developed in a simulation setting to assess the effects of AR cues, in the form of a bounding box, on the driver's awareness of pedestrians (DAP). Moreover, The DAP has been described using 3 levels of variables, consisting of the perception, vigilance, and anticipation. In the experiment setup, 27 participants tested the system in a fixed-base driving simulator. The participants were tasked to follow a lead vehicle, encountering scenarios in which pedestrians crossed the road. The dependent variables, such as visual reaction time, the accelerator pedal position, and the amount of urgent braking, were analyzed for each pedestrian situation. The results of the study revealed significant improvements in the dependent variables using the proposed AR cues for each level of the DAP, while not producing any negative masking effects.

Colley et al. [14] who compared four different modalities to indicate the object which may collide with the autonomous driving vehicle, also proposed a similar visualization. Two of them are the visualization on the windshield display, the rest two are HUD and lightbands on the dashboard. However, the result shows that there is no significant differences regarding the acceptance by participants.

**2.2.2 Visualizing dangerous areas/trajectories.** Instead of merely marking the hazardous object itself, many AR interfaces assist the driver in steering clear of perilous conditions by signaling that the area could be hazardous to enter. For example, Tonnis et al. [63] developed two visualization schemes to display the braking distance and drive-path using an AR HUD. As most car incidents occur due to longitudinal collisions or lane departure, the researchers assumed that providing the drivers with this type of information could be beneficial. The concept was evaluated in a fixed-base driving simulator, resulting in some interesting human factor findings, such as that increasing visual aid provided an increased feeling of safety and did not affect the overall workload. Another visualization shows the trajectory of other road users. For example, Colley et al. [15] proposed an AR interface on the windshield display that visually displays the future trajectory predicted of other vehicles, to improve the transparency of the autonomous driving system. Compared with 3 variables, color (hue), shape, and transparency, in the user study, the results show that color conveys uncertainty uncertainty best.

*In summary, the automotive research community has extensively investigated AR interfaces for driving tasks, including automated driving and driver assistance, with a particular focus on AR head-up displays and windshield displays. However, it remains unclear whether these designs can be effectively extended to AR head-mounted displays (AR-HMDs), as no prior research has examined interface design in this domain.*

### 2.3 Impact of AR on Driver Distraction

While the implementation of AR technology can enhance the cognitive efficiency of the driver [38], or reduce the mental workload [3, 4, 7, 49], potentially leading to a higher task performance [39], studies have also shown that it has its potential drawbacks, such as visual clutter or information overload [51]. For example, Horrey et al. [26] found that the increased clutter with the information overlay did not necessarily disrupt task performance more than a conventional adjacent display. It was suggested that the interference from visual clutter was not significant as the driving task is more reliant to the ambient vision. However, in more sophisticated AR systems that highlight potential hazards, it must be noted that interference can also be caused by the redirection of the user's gaze from critical road elements that are not augmented by the system [31].

Méndez et al. [47] conducted a review on the cognitive mechanisms of contemporary use of AR as ADAS. In the automotive industry, the implementation of AR systems is expected to aid the driver in navigation and hazard detection in order to reduce the amount of human factor related road accidents. However, the author suggests that there is a plausible prospect that the implemented system offers little help to the driver, while adding risk of cognitive and perceptual overload. It has been concluded that AR information should not be present at all times to prevent occlusion of road elements and over trust of the system. Another related implication in the field is inattentive blindness, which is a type of cognitive tunnelling, in which the person fails to perceive an event, as their attention is focused on another stimulus. The study by Chen et al. [12] revealed that inattentive blindness to unexpected hazard is enhanced by AR-HUD assisted driving. It has been found that unpredictable stimuli, accidentally superimposed on the AR graphic, lead to a higher probability of the driver ignoring the accidental events, and increased their response time. As can be frequently observed, fixed-base driving simulator studies on AR-HUD conducting a subjective measurement of the mental workload through a validated questionnaire such as the NASA-TLX or DALI, found a high standard deviation in combination with no statistical significance on self-reported mental workload [18, 31, 32, 36, 63].

In the research conducted by Kun et al. [33], participants were allowed to place the Skype window at a fixed position of their choice. The findings indicate that AR content is less distracting compared to a traditional monitor display. In another study by He et al. [23], the authors evaluated four different positions for navigation icons and discovered that positioning the icons in the center, directly in front of the windshield, caused the most distraction, despite being more accessible for participants.

*In summary, the design elements of such a system greatly determines the degree of the interference to visual attention or cognitive load. Compare with extensive research investigated the distraction of AR-HUD, the study using AR-HMD are rather sparse, especially there is no research using AR-HMD project holographic on the environment.*

## 3 Method

To gain deeper insights into the shape and impact of AR-HMD projections in driving contexts, two distinct AR designs were developed and tested across multiple driving scenarios. The research questions and hypotheses for this study were defined as follows:

- Q1 - What is the user's level of acceptance of AR-HMD driving assistance applications?
- Q2 - Which forms of holographic design do users prefer? How do different AR designs influence driving behavior, particularly in interactions with other road users?
- Q3 - what extent do AR-designated risk areas impact the driver's cognitive load?

### 3.1 Participants

The experiment has been conducted with 21 participants. Due to motion sickness occurring in 4 participants, the experiment had only been partially executed. Therefore, the data of these participants were excluded from the analysis. The mean age for the remaining participants (n=17) was 34.1 (SD = 6.6) and the gender distribution was



16 male and 1 female. Moreover, the mean years of driving experience was found to be 15.4 (SD = 7.3), and 14 participants were daily drivers. None of the participants have participated in a study of similar context before.

### 3.2 Apparatus

For the experiment, a fixed-base driving simulator has been developed, consisting of three 126cm diagonal LCD display panels, each with a resolution of 1920x1080. The simulator also includes a set of 500W surround sound speakers, a Logitech G27 steering wheel with pedals, and an adjustable driver's seat (See Fig. 1a).

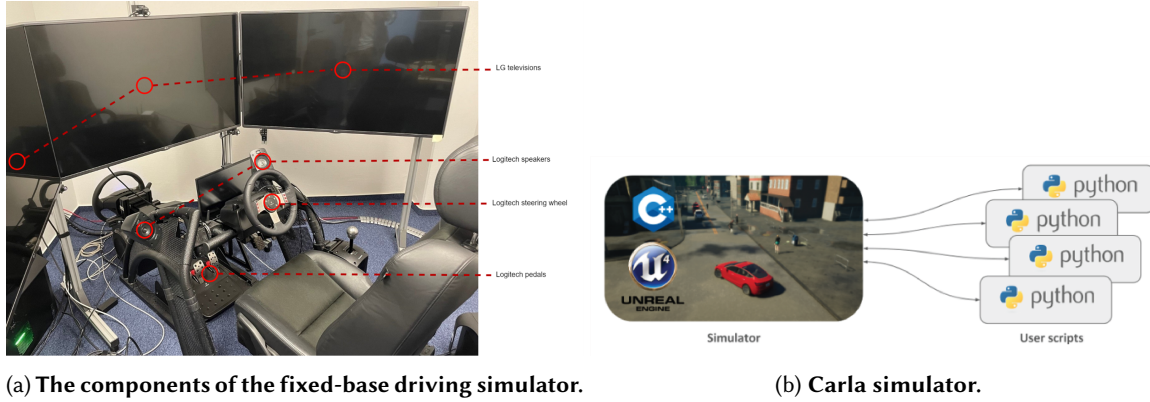


Fig. 1. The hardware and software of the fixed-base driving simulator.

The simulated environment is created using Carla (0.9.12), an open-source simulator for autonomous driving research (See Fig. 1b) [61]. In this project, the Python API of Carla has been modified to function as a driving simulator. Furthermore, the software is integrated with various compatible modules, including scenario runner and Carla ROS bridge. These integrations enable the execution of complex, repeatable test driving scenarios with real-time data communication.

Using the Unity platform [65] and the open-source MRTK library (v 2.7)[45], an AR application has been built to be deployed to a Hololens 2. The application receives real-time data from the driving simulator via ROS 1 [54], and performs the calculations in real-time to display the risk areas (See Fig. ??).

### 3.3 Risk Calculation Formula

In order to find a robust method to determine the timing and behaviour of the collision risk zone, the assumption has been made that the traffic objects at risk of collision are represented as points and move with a constant velocity vector in 2-D space.

A widely known time metric to quantify the risk of a future driving scenario is the time-to-collision (TTC). While the metric is successfully used by studies to solve 1-D problems, for example, vehicle following collisions or approaching stationary object collisions, the application in 2-D space is severely limited as the TTC assumes scalar quantities of vehicle motion [40, 73, 73]. As two traffic objects may not always follow trajectories that lead to a collision, it's not always feasible to calculate a value for the (TTC) to use as a basis for visualization. Therefore, an alternative formula has been derived that determines the time it takes for the two objects to reach their closest point to each other. This metric, The time to minimum distance (TTMD) is similar to the time to collision in approaching 1-D scenarios. However, does not assume scalar quantities and is therefore applicable in

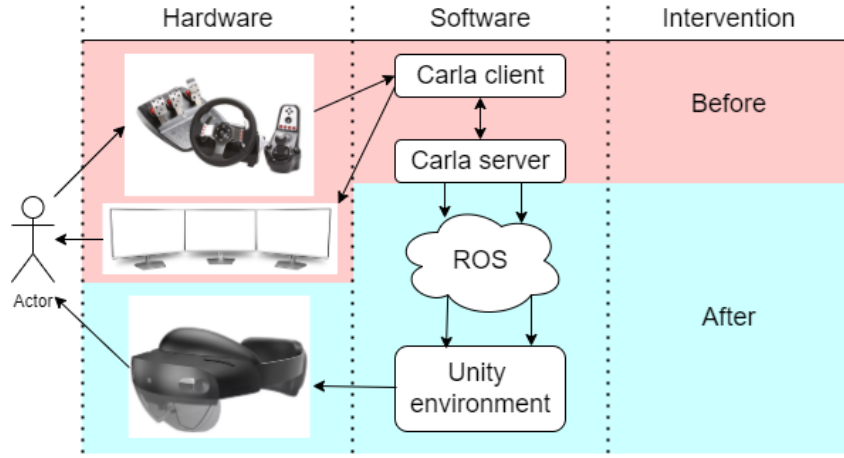


Fig. 2. The data infrastructure of the simulation setup.

2-D for any conflicting trajectories. The formula for the TTMD can be seen in equation 1.

$$t_{min} = -\frac{(\Delta \vec{p} \cdot \Delta \vec{v})}{(\Delta \vec{v} \cdot \Delta \vec{v})} \quad (1)$$

In which  $p$  is the absolute position vector ( $x, y, z$ ) in meters, and  $v$  is the velocity vector ( $x, y, z$ ) in meters per second. If  $t_{min}$  is less than zero, the closest point in time has already passed. On the contrary, if  $t_{min}$  is greater than zero, the time and distance to the closest point of approach can be calculated for both objects in order to find the degree of criticality to base the conditions of visualization upon. Given that the TTMD shares similarities with the TTC in 1-D traffic scenarios, the critical time values were approximated based on previous studies where the TTC was accepted as a surrogate measure for road safety [5, 8, 19, 41, 53, 57, 66]. For indication of a safety-critical situation, TTC values between 1.5 and 5 seconds are frequently found, for which 4 or 5 seconds is a recommended upper bound of activating a collision avoidance system [19, 41], and 3 seconds for distinguishing unsafe situations [5, 25, 53]. However, some researchers argue 1.5 seconds or even lower to be more desirable for classifying a critical event as false positives in a warning system may lead to other implications such as annoyance and trust in the system [8, 57, 66].

In this study, the threshold for visualizing a risk area is based on the concept of a near collision risk with a future colliding trajectory and generalized vehicle dimensions. This criterion is applied in situations where the minimum distance between an object and the ego vehicle is less than 5 meters, along with a TTMD value of less than 5 seconds. If both conditions are met, the risk area is visualized in accordance with the following cases:

- case 1 - if  $t_{min}$  is greater than 0 AND  $t_{min}$  is less than / equal to 2.
- case 2 - if  $t_{min}$  is greater than 2 AND if  $t_{min}$  is less than / equal to 5.

In case 1, the risk area is represented based on the concept of a near collision, calculated as the product of the relative velocity and the TTMD, as shown in equation 2.

$$L_{area} = \Delta \vec{v} * t_{min} \quad (2)$$

In case 2, the multiplication of the relative velocity with a factor 2 yields the distance corresponding to a 2 seconds TTMD, as shown in equation 3.

$$L_{area} = \Delta \vec{v} * 2 \quad (3)$$



As for the TTMD value distinguishing a critical and non-critical collision risk, a value of 2 seconds has been chosen, which is an approximate of the recommended TTC values from the previous studies [5, 8, 25, 53, 57, 66].

### 3.4 AR Risk Area Designs

It is challenging to enumerate and evaluate all design elements exhaustively. However, as discussed in Chapter 2.2, AR-enhanced driving warnings can generally be categorized into two types: visualizing directly on objects and highlighting hazardous areas. In this section, we present two representative designs that cover both categories for an initial examination of this topic, drawn from previous research by Wang et al. [69].

For implementation, the two designs are distinct to each other as they scale on a different plane in Euclidean 3D space. While design 1 scales strictly on the road plane (x,y), which highlighting the hazardous areas; design 2 scales strictly perpendicular to the road plane (z), which directly highlighting risky object. Furthermore, both designs can be displayed on the front or the back of the other vehicle, depending on the relative orientation and the direction of the collision risk. Finally, A medium opacity red color has been used for both designs to indicate danger in traffic, while minimizing the visual clutter that blocks the vision of the driver.

**3.4.1 Red carpet - Design 1.** In the scenario of following a lead vehicle that suddenly decelerates, leading to a TTMD of less than 2 seconds, a red transparent carpet is displayed on the road behind the lead vehicle (See Fig. 3).

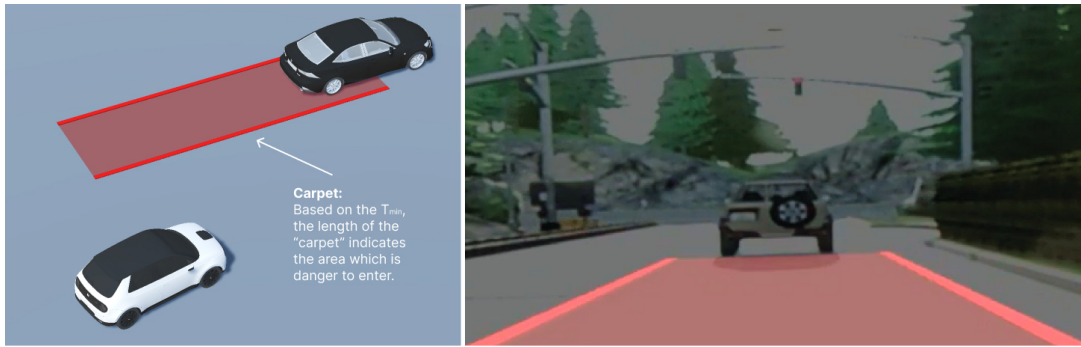


Fig. 3. Design 1 from a participant's perspective.

The length of the carpet is based on the difference in speed multiplied by the time to minimum distance. In case the TTMD is between 2 and 5 seconds, the length of the risk area starting from the back of the lead vehicle covers a road area corresponding to a 2 seconds TTMD. This results in a distinguishable border between the areas of imminent and non-imminent danger. The visualization is integrated according to the following formula:

$$L_{area} = \begin{cases} \Delta v * t_{min\_distance}, & \text{if } t_{min\_distance} \leq 2 \\ \Delta v * 2, & \text{otherwise} \end{cases} \quad (4)$$

**3.4.2 Virtual wall - Design 2.** In the scenario of following a lead vehicle that suddenly decelerates, a red wall consisting of transparent red blocks layered on top of each other, is displayed on the back of the lead vehicle (See Fig. 4).



Fig. 4. Design 2 from a participant's perspective.

The length of the wall increases until the height of the car as the time to minimum distance goes to 0. Moreover, the transparency of the individual blocks are higher at the top layers to prevent obstructing the view of the vehicle. This feature is added on purpose, as the rear window could present crucial information, for example, law enforcement LED sign, braking light, or the road situation in front of the vehicle. The visualization is integrated according to the following formula:

$$H_w = \begin{cases} h_{obj} * (t_{min\_distance}/2), & \text{if } t_{min\_distance} \leq 2 \\ h_{obj}, & \text{otherwise} \end{cases} \quad (5)$$

In which  $h_{obj}$  is the object's (car or pedestrian) height.

### 3.5 Experiment Design

In this study, the experiment was setup as two separate 3 x 1 within-subjects factorial designs. The AR projections (Design 1, Design 2, and Baseline) were considered as the independent variables in each of the towns (A or B) (See Table 1 & Fig. 5).

Table 1. Experiment sessions and conditions.

| Session | Condition | Town |
|---------|-----------|------|
| 1       | Baseline  | A    |
| 2       | Design 1  | B    |
| 3       | Design 1  | A    |
| 4       | Design 2  | B    |
| 5       | Design 2  | A    |
| 6       | Baseline  | B    |

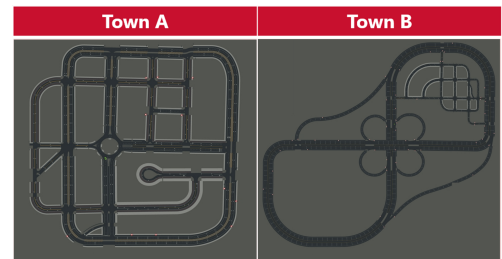


Fig. 5. Experiment town layouts.

The towns are maintained as distinct variables because the primary aim of the experiment does not involve comparing or contrasting them. However, given the different characteristics of the two towns, with Town A situated within an urban environment, while Town B resided within a suburban landscape, the layout of the towns may influence the participant's driving behaviour and user experience.

In Town A, the route included more elements, such as denser buildings and billboards, that led to a higher rate of optic flow and more 90-degree turns, which can increase the likelihood of simulator sickness [24] [55].

On the other hand, Town B featured a significant change in velocity, as participants transitioned from highway driving to a complete standstill at a traffic light after a sharp curve, which could also induce simulator sickness. To address the potential issues of these factors, participants were questioned after each drive, and the experiment was halted if any symptoms of motion sickness were observed.

Since the concept requires the driver to wear a HMD to view the AR projections, additional influences such as physical discomfort or visual distraction may impact the driving behaviour or cognitive load. Therefore, as a baseline, a no design condition has been introduced in the experiment. In this condition, the driver is still required to wear the HMD which displays no AR risk areas. The only remaining visible AR projection is the speedometer, which is also present in the other conditions.

Over the course of a driving simulator study, participants may become more accustomed to the simulator or use case scenarios through repeated exposure. Therefore, to account for this learning effect, a second baseline session was introduced as the last session of the experiment to assess whether the observed changes are due to learning effects or introduction of AR risk area projections. Moreover, in order to further diminish the potential for learning effects, the driving sessions were alternated between towns A and B, thereby increasing the challenge of memorizing the route and traffic scenarios. The order that was presented to the drivers can be seen in table 1. In total, 6 sets of simulator data and 6 questionnaires were collected for each combination of AR condition (Design 1, Design 2, and Baseline) and Town (A or B).

### 3.6 Procedure

The experiment consisted of 6 driving sessions lasting a total of 65 minutes (See Appendix). Before the start of the driving sessions, the participants had to read and sign an informed consent form. Then, a briefing was held in which they received information regarding the procedure of the experiment and the driving task. As the participants were invited in the simulator chair and equipped with the Microsoft HoloLens2, the participants were also given the option to adjust the seat of the driving simulator for a more comfortable driving position. Afterwards, a training session was initiated for which they are instructed to follow a lead vehicle for 3 minutes in a town with no other traffic participants. Here, a reminder is given that in case of nausea, the participant must inform the researcher to halt the experiment. This session was intended for the participants to familiarize with the operation of the driving simulator and following a lead vehicle. (See Fig. 6).

Upon completion of the training session, the participants were asked on their well-being and questions left. Then, for each of the driving sessions, the driving simulator and the Microsoft HoloLens 2 were configured to start with the corresponding AR condition and town. The participants were given the instructions to follow the lead vehicle while respecting all the traffic rules. During the driving sessions, no random traffic participants were present in the town. Only vehicles and pedestrians that were part of the scenarios were triggered to spawn outside the view of ego vehicle, and moved inside the view of ego vehicle according to the scenarios. After each session, the participants filled out a questionnaire. After all the driving sessions, a short interview was held in which participants could provide verbal feedback about the experiment.

### 3.7 Scenarios

From the Carla repository, two different towns (See Fig. 5) have been chosen to implement the following traffic scenarios:

- Town A - Unsignalized intersection (2x), sudden deceleration, roundabout
- Town B - Sudden deceleration, approaching slower vehicle, pedestrian crossing, cut-in

The towns exhibit contrasting characteristics, with Town A situated within an urban environment characterized by tall buildings, while Town B resides within a suburban landscape consisting of houses and a connecting highway. The different layout allows the implementation of specific scenarios in which the concept could assist



Fig. 6. A participant operating the system.

the driver in gauging the other traffic participant's acceleration and the risk of collision by displaying the risk area dynamically.

The following 6 different traffic scenarios (See Fig. 7) were developed:

- Unsignalized intersection (See Fig. 7a) - When the ego vehicle approaches an unsignalized intersection, the lead vehicle has already passed this intersection, and another vehicle simultaneously approaches from either the right or the left side, for which the vehicle coming from the right has priority. In case the ego vehicle has priority, the projected risk area can be seen decreasing in size, as the other vehicle decelerates to yield. In case the other vehicle has priority, the projected risk area can be seen remaining the same in size, indicating a constant velocity and no intention to slow down. From the size and superimposed road space of the risk area, the driver can now make a decision with this additional information whether it is safe to cross the intersection.
- Roundabout (See Fig. 7b) - In the scenario of the ego vehicle approaching a common roundabout, another vehicle enters the roundabout from the left, for which the vehicle on the roundabout has priority. Similar to the intersection scenario, the superimposed road space can assist the driver in deciding whether entering the roundabout may lead to a near collision.
- Sudden deceleration (See Fig. 7c) - A traffic light transitions to yellow, as the lead vehicle approaches. This causes the lead vehicle to suddenly decelerate in front of the ego vehicle, which triggers the risk area to quickly expand in size. This behaviour acts as a salient collision warning, and grants information regarding the intensity of the deceleration. Thus, assisting the driver to respond accordingly.
- Pedestrian crossing (See Fig. 7d) - A pedestrian crosses the road at a stop sign when approached by ego vehicle. The behaviour of the risk area is similar to the intersection scenario, as traffic participants are treated equally. However, a pedestrian is generally harder to notice than a vehicle, for which the risk area enhances the visibility, and additionally assists the driver in spotting the pedestrian.

- Approaching slower vehicle (See Fig. 7e) - The ego vehicle encounters a significantly slower vehicle on the highway. Even when both vehicles are moving with constant velocity, the significant size of the risk area can indicate a dangerous situation ahead.
- Cut-in (See Fig. 7f) - As the ego vehicle approaches two vehicles in sequence on the highway, an overtake maneuver ensues, for which the vehicle trying to overtake, cuts in front of the ego vehicle. This significant lateral movement triggers the risk area to superimpose onto the road ahead of the ego vehicle, which assists the driver in noticing the maneuver, and granting information regarding the potential of a collision.

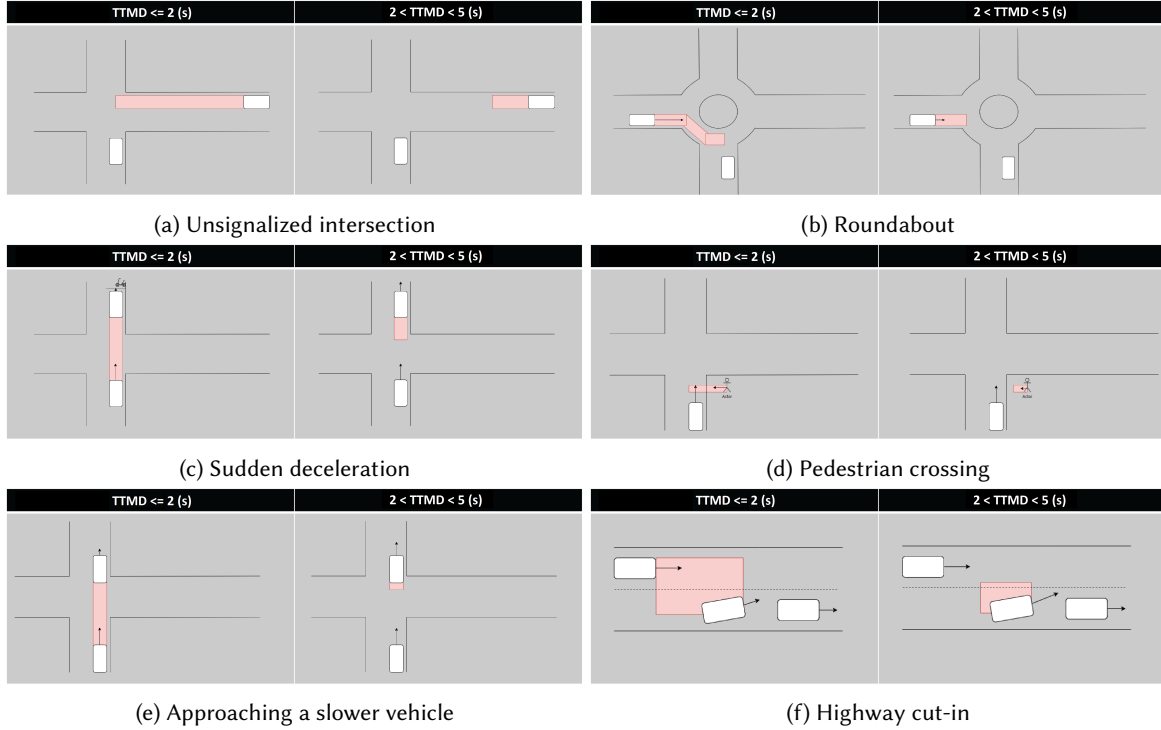


Fig. 7. Scenario graphics.

### 3.8 Data Collection and Variables

During the experiment, raw simulator data of all vehicle positions (x,y,z) were recorded at a frequency of 40 Hz to ROS bags. Following each driving session, a questionnaire was completed, and the responses were digitally recorded with a unique ID and session number. By utilizing the vehicle position data, the variables related to the driving behaviour and road safety, for example., velocity, acceleration, and distance between the vehicles, were calculated offline using MATLAB [42].

This study examines driving behaviour and road safety through several key variables, including time headway (THW), time-to-collision (TTC), brake response time, maximum deceleration, and yield success.

Time headway is defined as the duration that passes between the moment the front of the leading vehicle crosses a specific spot on the road and the moment the front of the trailing vehicle crosses that same spot, while the TTC represents the duration remaining until a collision takes place, assuming both vehicles persist

on their current paths and maintain their existing speeds [67]. The time headway reflects the potential risk of a collision, while the TTC indicates the actual risk of a collision. Only vehicles traveling with short headways can produce low TTC values, and in situations with low TTC, avoiding a collision requires proactive intervention [67]. Moreover, The TTC is further divided into the lowest value observed during a sudden braking event to assess severity, and the number of instances where TTC fell below 3 seconds throughout the driving session to measure the frequency of safety-critical moments [5, 25, 53]. Additionally, the study takes into account the brake response time and the maximum deceleration, which are determined by the first moment the driver engages the brake pedal following a sudden deceleration of the lead vehicle, to indicate the criticality [1, 21, 46]. By introducing the scenario of unsignalized intersections where the driver must yield, the inclusion of the yield success served as an indication of the driver's ability to assess the speed and distance of oncoming vehicles.

### 3.9 Analysis

The dependent measures consisting of the driving simulator data (THW, TTC, Brake response time, and Maximum deceleration), were analyzed using repeated measures ANOVA. However, when normality was violated according to the Shapiro-Wilk (S-W) test, the Friedman test was employed as an alternative. As the yield success was represented as a binary value (1=yield, 0=no yield), Cochran's Q test was used to analyze the results. For all the reported results, an  $\alpha$  level of 0.05 was used to determine the statistical significance. Post-hoc tests, adjusted with the Bonferroni correction, were conducted for pairwise comparisons, for example, Baseline vs. Design 1, Baseline vs. Design 2, Design 1 vs. Design 2. The  $\alpha$  level in this case, with Bonferroni adjustment, was .017.

To evaluate the subjective responses, which were based on 7-point Likert scale ratings regarding the cognitive load and user experience, pairwise comparisons were conducted using the Wilcoxon signed-rank test. The significance level ( $\alpha$ ) was set at 0.017 after applying the Bonferroni adjustment.

## 4 Results

### 4.1 Time Headway

For each driving session, the time headway was measured as indication for the degree of safe distance keeping. In the US, several driver training programs [17] suggest that following a vehicle with a headway of less than 2 seconds is unsafe. To analyze this, the duration (in seconds) of the time headway below 2 seconds was calculated for the whole driving session and during the abrupt braking scenario to assess the impact of each condition. The duration of the time headway below 2 seconds is represented on the Y-axis, while each condition is plotted on the X-axis in Figure 8

As the assumption of normality was only violated in Town A, a Friedman's test was performed for the data of Town A, while a repeated measures ANOVA was used for Town B. The results indicated no significant differences in the duration of the time headway below 2 seconds among the conditions in Town A ( $X^2(2) = 0.57$ ,  $p > 0.05$ ) and in Town B ( $F(2,32) = 0.08$ ,  $p > 0.05$ ). Similarly, during abrupt braking scenarios, no significant differences were found in Town A ( $X^2(2) = 0.35$ ,  $p > 0.05$ ) and in Town B ( $F(2,32) = 0.43$ ,  $p > 0.05$ ).

Two Friedman's tests were conducted to analyze the mean time headway (in seconds) during abrupt braking scenarios. In Town B, a significant difference was found between at least two conditions ( $X^2(2) = 7.42$ ,  $p < 0.03$ ). However, in Town A, there was no significant difference ( $X^2(2) = 0.48$ ,  $p > 0.05$ ) among the conditions. An overview of the box plots can be seen in 9.

A post-hoc pairwise comparison revealed that the baseline differed significantly from Design 1 ( $p < .005$ ) and Design 2 ( $p < .009$ ). Additionally, Design 1 and Design 2 showed no significant difference ( $p > .017$ ). These results suggest that in scenarios involving sudden braking without AR projections, drivers tended to maintain a shorter following distance behind the lead vehicle compared to situations where AR projections were displayed.



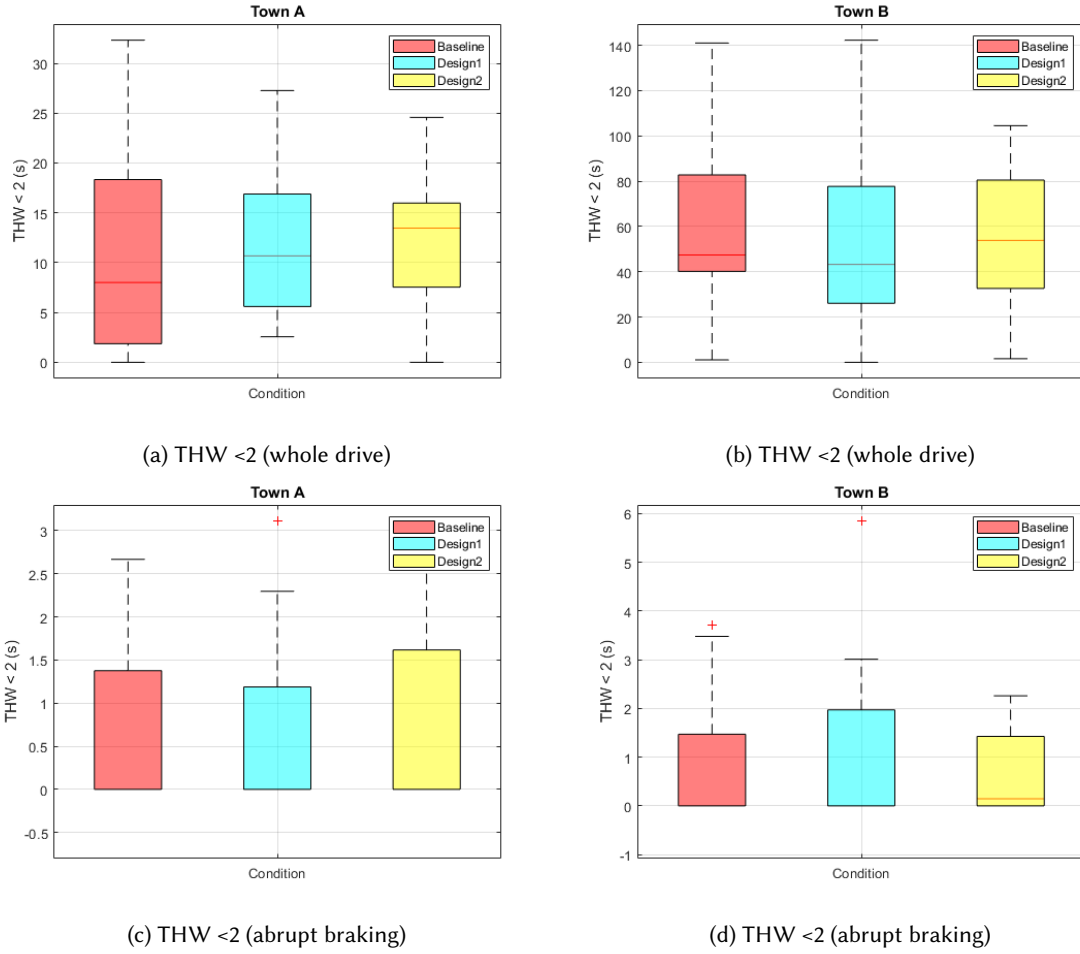


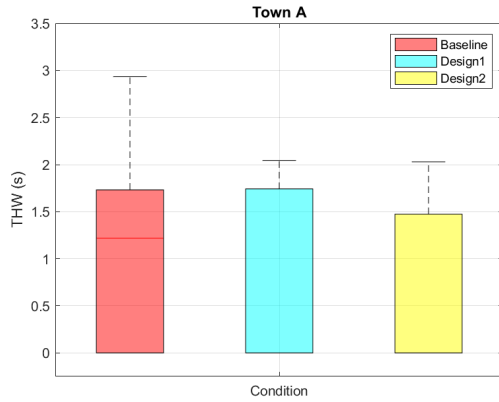
Fig. 8. The duration of THW < 2s for the whole driving session and abrupt braking scenario.

## 4.2 TTC

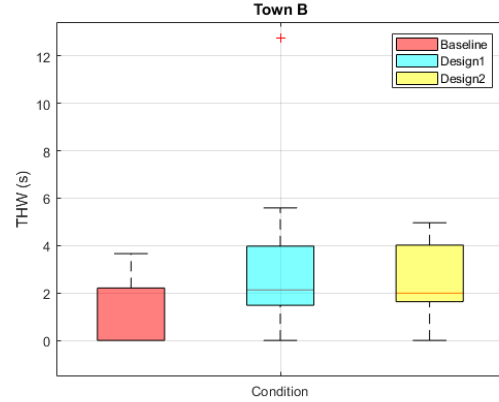
**4.2.1 Minimum TTC.** The minimum TTC is the lowest value observed during a sudden braking event, as shown in Figure 10.

A repeated measures ANOVA test was conducted on the minimum TTC in town A, but no significant differences were found ( $F(2,32) = .14, p > .05$ ). In town B, where the assumption of normality was violated according to the Shapiro-Wilk test, an alternative Friedman's test was performed, and it also revealed no differences ( $\chi^2(2) = .47, p > .05$ ). This finding indicates that AR projections did not lead to a notable impact on the minimum TTC when compared to no AR projections.

**4.2.2 TTC below 3 seconds.** Since many ADAS systems initiate Automatic Emergency Braking (AEB) at a TTC below 2 seconds and provide warnings at 3 seconds [16, 58], we also consider 3 seconds as the threshold for dangerous proximity. The TTC below 3 seconds represents the number of instances where the TTC fell below 3 seconds during the driving session (see Fig. 11).

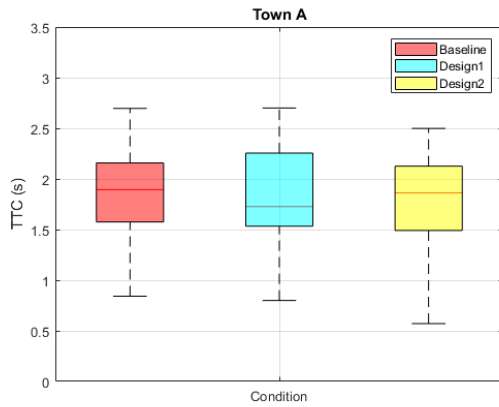


(a) Mean THW (abrupt braking)

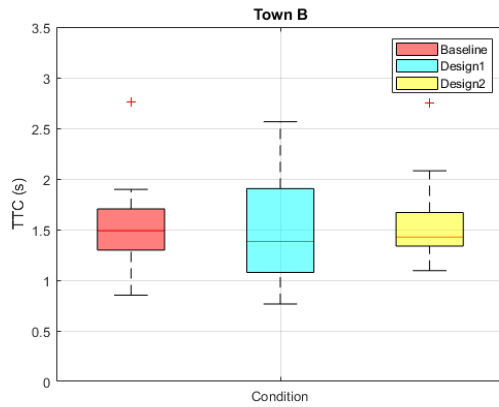


(b) Mean THW (abrupt braking)

Fig. 9. The mean time headway during abrupt braking scenario.



(a) Minimum TTC (abrupt braking)



(b) Minimum TTC (abrupt braking)

Fig. 10. The minimum TTC during abrupt braking scenario.

By conducting a Friedman's test for both towns, no differences were observed in either town (Town A:  $\chi^2(2) = 2.21$ ,  $p > .05$ ; Town B:  $\chi^2(2) = 1.09$ ,  $p > .05$ ). These findings indicate that the AR projections had no significant impact on the number of TTC occurrences below 3 seconds in either town.

#### 4.3 Brake response time

The brake response time of the driver during abrupt braking scenarios was determined by calculating the time difference between the lead vehicle braking and the driver pressing the brake pedal. A repeated measures ANOVA test was conducted on the brake response time, and the results showed no significant differences in either town

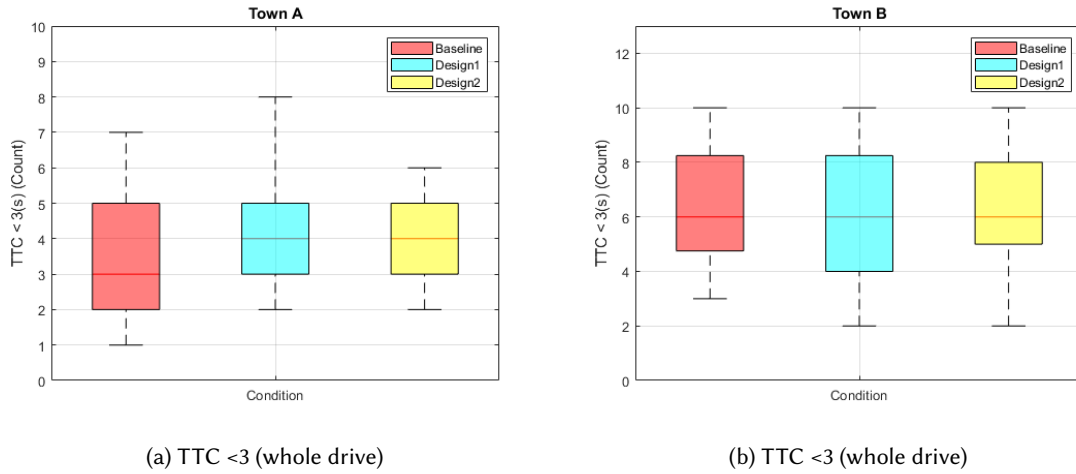
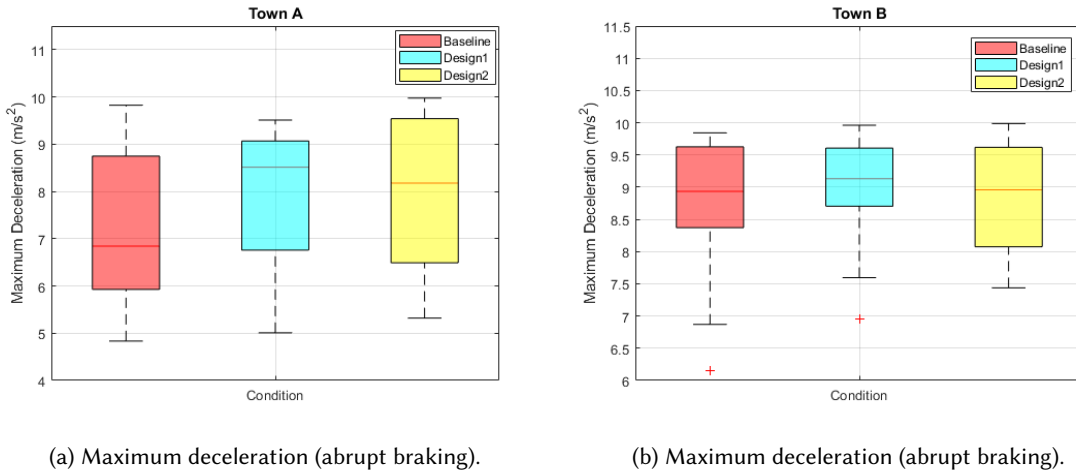


Fig. 11. The occurrences of TTC &lt; 3 seconds during the driving session.

(Town A:  $F(2, 32) = .24, p > .05$ ; Town B:  $F(2, 32) = .14, p > .05$ ). This indicates that the AR projections did not influence the response time of the driver at the event of an abrupt deceleration of the lead vehicle.

#### 4.4 Maximum Deceleration

The maximum deceleration ( $m/s^2$ ) is the highest value observed during a sudden braking event (See Fig. 12).

Fig. 12. The maximum deceleration ( $m/s^2$ ) during abrupt braking.

Since the S-W test indicated a violation of the normality assumption, the alternative Friedman's test was conducted for both Town A and Town B. The results of the test showed no significant differences in terms of the impact of AR projections on the maximum deceleration in either town (Town A:  $\chi^2(2) = .82, p > .05$ ; Town B:  $\chi^2(2) =$

.47,  $p > .05$ ). This suggests that the AR projections did not have an effect on the maximum deceleration during a sudden braking event.

#### 4.5 Unsignalized Intersection Yield Success

The yield success is a binary representation whether the driver yielded (1) or did not yield (0). As the unsignalized intersection scenario, in which the driver is required to yield, appeared twice, and at different locations, the appearances were treated independently, for example, appearance 1 and appearance 2. By performing a Cochran's Q test, there was a statistically significant difference in yield success between the conditions in the first appearance ( $Q(2) = 7.40$ ,  $p < .03$ ). However, not in the second appearance ( $Q(2) = 1.14$ ,  $p > .05$ ).

In a post-hoc pairwise comparison between the conditions, it was found that Design 2 differed significantly from the baseline ( $p < .001$ ), in terms of yielding behavior. However, there was no significant difference between Design 2 and Design 1 ( $p > .017$ ). This suggests that drivers were more likely to yield when influenced by the AR projections from Design 2 compared to scenarios with no AR projections at all (baseline).

#### 4.6 Questionnaire

The participants were prompted to rate a set of questions cited from the validated user experience questionnaire (UEQ) [34] and the driving activity load index (DALI) [50] using a 7-point Likert-type scale, with 1 = "Extremely low", 2 = "Very low", 3 = "Low", 4 = "Neutral", 5 = "High", 6 = "Very high", and 7 = "Extremely high", after each test drive in the simulation. To statistically analyze the subjective response for significant differences, a Wilcoxon signed-rank test has been performed on the questionnaire data, and a Bonferroni correction has been applied for which a new alpha level ( $\alpha = 0.05/3$ ) was found to determine statistical significance. The results for which the null hypothesis could be rejected ( $p < .0167$ ) can be seen in table 13f.

From the mean scores of the subjective response for which the participant could rate the attractiveness of the system from 1 (very low) to 4 (neutral) to 7 (very high), it appears that Design 1 ( $\mu_A = 4.71$ ,  $\mu_B = 4.59$ ) is valued higher in attractiveness compared to Design 2 ( $\mu_A = 4.00$ ,  $\mu_B = 3.59$ ), and the baseline ( $\mu_A = 3.82$ ,  $\mu_B = 3.82$ ) (See Table 13a & Fig. 13b & Fig. 13c). However, the Wilcoxon signed-rank test revealed only statistically significant differences between Design 1 and Design 2 ( $p < .0167$ ) (See Table 13f).

When participants could rate the amount of interference caused by the AR projections from 1 (very low) to 4 (neutral) to 7 (very high), it has been found that Design 2 ( $\mu_A = 3.76$ ,  $\mu_B = 3.71$ ) had a larger negative impact on the driving task compared to Design 1 ( $\mu_A = 3.06$ ,  $\mu_B = 3.59$ ) and the Baseline ( $\mu_A = 2.41$ ,  $\mu_B = 3.06$ ) (See Table 13a & Fig. 13d & Fig. 13e). The Wilcoxon signed-rank test revealed statistically significant differences between Design 2 and the Baseline in both towns, and between Design 2 and Design 1 only in town A ( $p < .0167$ ) (See Table 13f). This finding indicates a difference between the two towns regarding the perceived interference.

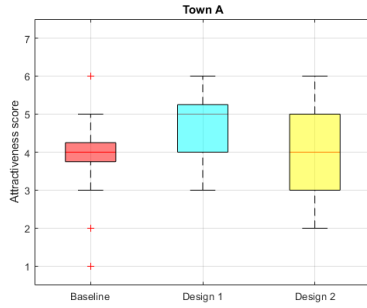
#### 4.7 Qualitative Feedback

At the end of the experiment, additional insights on the participant's subjective rationale were collected through semi-structured interviews. By conducting an inductive thematic analysis approach, various themes were generated by clustering the common codes among the participants.

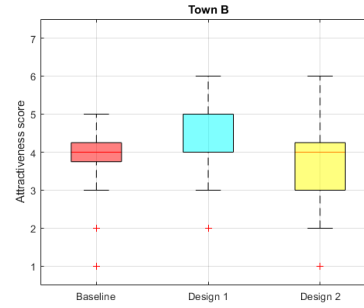
**4.7.1 Feedback on the risk visualization design elements.** As the purpose of the concept is to provide the driver with additional information regarding a risk of collision, the design elements of the augmented reality visuals were carefully chosen to minimize the interference on the driving task. About half of the participants mentioned a mismatch in the visualization of the risk area and the urgency of the situation. As the size of the risk area was directly derived from the relative orientation and speed of the vehicles, the participants could have experienced heavily changing risk areas when driving with inconsistent speeds. "The change of risk area is too abrupt." "When the car on the right came on the right, the risk area popped up, which startled me. I feel like braking immediately,

|        | Interference                | Attractiveness              |
|--------|-----------------------------|-----------------------------|
| B (A)  | $\mu = 2.41, \sigma = 1.06$ | $\mu = 3.82, \sigma = 1.24$ |
| D1 (A) | $\mu = 3.06, \sigma = 1.14$ | $\mu = 4.71, \sigma = 0.99$ |
| D2 (A) | $\mu = 3.76, \sigma = 1.15$ | $\mu = 4.00, \sigma = 1.17$ |
| B (B)  | $\mu = 3.06, \sigma = 1.56$ | $\mu = 3.82, \sigma = 1.07$ |
| D1 (B) | $\mu = 3.59, \sigma = 1.28$ | $\mu = 4.59, \sigma = 1.06$ |
| D2 (B) | $\mu = 3.71, \sigma = 1.05$ | $\mu = 3.59, \sigma = 1.28$ |

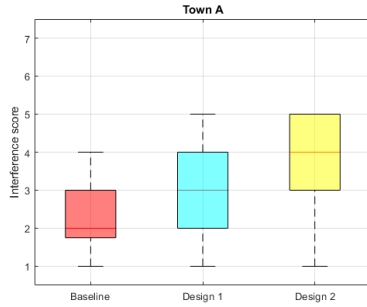
(a) Mean and standard deviation scores of the attractiveness and interference rating, categorized by condition and scenario town.



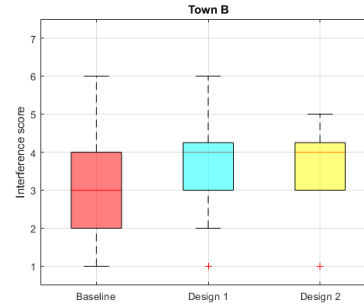
(b) Attractiveness score in town A.



(c) Attractiveness score in town B.



(d) Interference score in town A.



(e) Interference score in town B.

|             | Interference      | Attractiveness    |
|-------------|-------------------|-------------------|
| B - D1 (A)  | NS                | NS                |
| B - D2 (A)  | 0.0549            | NS                |
| D1 - D2 (A) | 0.013671875000000 | 0.015625000000000 |
| B - D1 (B)  | NS                | NS                |
| B - D2 (B)  | 0.159179687500000 | NS                |
| D1 - D2 (B) | NS                | 0.011230468750000 |
| B - B (A-B) | NS                | NS                |

(f) Pairwise comparison between the different conditions and driving scenarios.

Fig. 13. Results of the questionnaire data. (B = Baseline, D1 = Design 1, D2 = Design 2, (A) = Town A, (B) = Town B)

and that might lead to a collision from behind." Furthermore, some of these participants have also mentioned the static red colour to be more fitting to imminent or urgent situations. "The red area makes me feel like stopping immediately." This indicates that the chosen colour and intensity might be too strong of a message for non-critical situations.

**4.7.2 Feedback on both designs.** When comparing Design 1 and Design 2, it's apparent that the participants generally preferred Design 1 over the baseline and Design 2. The majority of the participants (16 out of 21) expressed a preference for Design 1, mainly due to its feature of the changing risk area on the horizontal plane. This aspect was highly regarded as they mentioned that it aligned between the car's movement direction and the risk area's movement direction. Participants also appreciated that Design 1 offered a spatial extension, providing more informative feedback. One participant pointed out that "You don't need to focus on the vehicle to see there is a vehicle." Additionally, Design 1 was seen as beneficial for maintaining safe distances, with a participant noting, "I know how much I can accelerate and brake and would not collide. It was very helpful to keep the distance."

On the other hand, a smaller group of participants (3 out of 21) favored Design 2, primarily citing its advantage of being visually less cluttered. Only one participant mentioned a preference for both designs, depending on the specific situation. They noted that "For intersecting trajectories, the line redirects the user's attention to the risk of collision." However, for scenarios involving cut-ins or sudden deceleration, the potential collision point was already apparent, and Design 2, with its less intrusive presentation, was considered more suitable.

In summary, from a general standpoint, Design 1 appears to be the preferred choice due to its intuitiveness and functional benefits. It eliminates the mismatch between the moving direction of the risk area and the the dimensional plane, and provides more salient information regarding the other vehicle's acceleration, which resonated positively with the majority of the participants.

## 5 Discussion

In this section, both the qualitative and quantitative results and limitations of the user study are discussed.

### 5.1 Effects of AR Risk Visualization on the Driving Behaviour

The data suggests that the introduction of AR risk visualizations during driving encourages safer driving behaviour, as the time headway between the ego vehicle and the lead vehicle during abrupt braking is higher than the time headway in case no projections are shown. However, the analysis of the response time, TTC, and maximum deceleration during abrupt braking scenarios did not reveal differences between the conditions. Therefore, the results suggest that the AR projections as shown in this study may not significantly contribute to enhanced road safety during critical events like abrupt braking.

It's worth noting that some previous studies, such as Calvi et al. [9] and Kim et al. [32], reported a significant decrease in reaction time when AR warnings were displayed in the case of abrupt braking scenarios. The main difference between these AR projections and those in this study is that in these studies, the projections were only displayed when the lead vehicle starts braking, or slightly before. In contrast, the AR projections in this study were triggered when the TTMD was below 5 seconds, which could lead to more frequent occurrences if the participant drove more aggressively. This repeated exposure might have contributed to a diminished response to the stimulus. However, further research is needed to validate this potential effect.

In a manner similar to the study by Phan et al. [52], the projection of AR risk areas might indirectly highlight hazardous objects, potentially leading to an increase in yield success. This increase could be linked to heightened situational awareness resulting from the redirection of the driver's attention toward potential risks. The data from this study suggests that the AR design supports the driving task by highlighting a potential collision risk to the driver, as shown by the increased amount of required unsignalized intersection yielding. This finding aligns with several other studies using AR cues to enhance the driver awareness [10, 52, 56].



## 5.2 Effects of AR Risk Visualization on Distraction and Cognitive Load

So far the results did not indicate a concern for safety related to the addition of visual clutter from AR projections or its impact on cognitive load. This is in contrast to a study by Tran et al. [64], where they designed a left-turn driving aid using a static red path AR projection in front of the approaching vehicle. Participants in their study reported that processing the path demanded a higher level of cognitive effort. Another study by Tonniss et al. (2007) [63], proposed a more simplistic design of a bar projection in front of ego vehicle to aid the driver in longitudinal and lateral driving. In their study, it was concluded that such a visualization did not affect overall workload. It's possible that factors such as the static length or opacity of the projection had a notable impact on the driver's cognitive load in those cases.

## 5.3 Subjective Response

Overall, the attractiveness of the driving system seemed to be perceived higher with the presence of Design 1 when compared to Design 2 or no AR projections. When comparing Design 2 to no AR projections, there was no clear preference for either condition. From observing the length of the box plots (See Fig. 13b & Fig. 13c) it can be suggested that the opinions were more divided in this condition. Since there was a strong preference for Design 1 compared to Design 2, as revealed by the differences found in the questionnaire data, the order effect (Design 2 after Design 1) may have lead to a contrast effect that influenced Design 2 to be rated as significantly less attractive than Design 1, instead of being rated independently. Based on the qualitative data, it might be that the difference in attractiveness between the two designs is associated with their functionality and intuitiveness. As from the interviews, some participants reported that the change of size of the AR projection in the horizontal plane felt more intuitive, as it was on the same plane as the moving direction of the vehicles. Therefore, participants' perceptions of the designs may have been influenced by how easily they could comprehend and interact with the system's features.

However, the questionnaire data indicates an increase in perceived interference in both towns when comparing AR projections of Design 2 to Design 1 and the baseline. This suggests that certain unique design elements of Design 2, for example, scaling direction, location relative to the traffic object, or transparency, may have strongly contributed to this impact. As found in a previous study by Rusch et al. [56], the AR cues in this study did not reveal any negative outcomes associated to interference, for which a similarity to Design 1 is that the AR cue does not obstruct the view of the traffic object. Nonetheless, additional research is necessary to thoroughly investigate the specific factors that contribute to this interference.

## 5.4 Limitations

Although this research yields valuable insights, several factors may have influenced the findings and should be considered when interpreting the results. The limitations discussed in the following sections concern both the experimental methodology and the use of the driving simulator.

**5.4.1 Experimental.** It's important to consider the number of critical events encountered during the simulation sessions in comparison to normal driving. The increased frequency of critical events in the controlled environment of the study might have influenced the participants' behaviour and reactions. When participants encountered these events sequentially, they may have been more attentive than they would be in real-world driving situations. Consequently, the potential benefits of the AR projections, such as indicating significant speed differences or sudden deceleration, could have been diminished. Furthermore, since all pedestrians and vehicles in the driving sessions were always part of a predefined scenario, this effect could have been amplified, as the driver can anticipate a critical event.

Another limitation for the sake of consistency and validity is the repetitive nature of the scenarios and the lack of counterbalance or randomization in the experiment design. While alternating between the towns may have

reduced habituation, the order effects were not accounted for. Therefore, in the case of yield success, Design 2 may have performed better than the other conditions as the participant may have recognized the scenario and responded accordingly. Furthermore, since the effect was only found in the first appearance of the traffic scenario, the different characteristics of the scenario might have influenced the yield success.

In the current setup, AR projections are not compared to other modalities, such as auditory and tactile. Although the primary purpose of projecting AR risk zones was not to warn the driver in critical events, but to inform the driver in advance of potential critical events, it would still be valuable to compare the effectiveness of AR as a warning system against other modalities in an additional control condition. In this manner, the strengths and weaknesses of AR projections could be assessed to validate the benefits of AR projections to other warning modalities.

During the study, participants completed six driving sessions in two different towns, in the same conditions. To mitigate the potential issue of participants recognizing scenarios, various alterations were made to the vehicles used in each driving session, including changes in brand and color. Despite these countermeasures, it's possible that some participants may have still recognized certain scenarios, such as the sudden braking event. This recognition could potentially affect the accuracy of the findings regarding the effects of the AR risk projections. However, the final driving session, which served as the second baseline session, did not reveal any significant results indicating participants' anticipation of upcoming scenarios.

Since the developed driving simulation environment limited the consistency of the simulation data by allowing the driver to freely maneuver, the dependent variables could be influenced by the difficulty of the driving task, leading to irregular following distance for instance. An included countermeasure for this behaviour was to include checkpoints in which the lead vehicle may wait at certain points until the ego vehicle is within a certain distance. However, between these checkpoints, there is still room for varying distance in which scenarios occur. A solution to this is to prevent the vehicles from drifting too far apart by controlling the acceleration of the lead vehicle, based on the distance ahead of the ego vehicle.

**5.4.2 driving simulator.** Regarding the handling of the driving simulator, it was reported that the difficulty of operating the driving simulator differed. The participants required various amount of time to execute the driving tasks on the machine on an adequate level. Moreover, Participants (8/21) mentioned: "It lacked smoothness. The frame rate was not consistent". Also the design of the urban driving scenarios included a substantial amount of 90 degrees cornering, which was found to be difficult to perform, as mentioned by 5 participants. However, most participants found it acceptable and easier as more time has passed in the experiment. This leads to similar statements such as "In the beginning it was difficult, at the end it gets easier." "You had to get used to the physics, then it was straightforward." "After a first couple of tries it wasn't hard." This effect over time may have influenced the driving behaviour as the participants that found it difficult in the beginning, may have been distracted by learning the driving simulator. However, comparisons between both baseline sessions (first and last session) revealed no differences.

Furthermore, the participants' proficiency of handling the driving simulator was divided, as their feedback on handling the driving task ranged from easy to difficult. The maneuver of executing a 90 degrees turn in an urban driving scenario was frequently mentioned to be one of the more difficult tasks, due to a lack of dynamic forces which would have been felt when driving a real car. However, the benefit of the chosen research method is an unconstrained driving scenario, leading to a more sophisticated driving experience for the participant. Thus, providing a better understanding of the concept in the real world. Perhaps with more training in prior of testing, or in case of a higher fidelity driving simulator, which includes dynamic feedback on the driver, the difficulty of the driving task may be diminished for more consistent results.

Due to technical limitations, it was not feasible to add brake lights in the simulation. When there are no brake lights, the driver could mainly notice a deceleration event by the narrowing gap between the vehicles. In the case

of abrupt braking, an expanding risk area has a higher salience than a narrowing gap between the vehicles for which the effects of the AR projections could be enhanced when comparing to a baseline condition with no brake lights.

Finally, a common side effect of a driving simulator is simulation sickness or motion sickness caused by the visuo-vestibular conflict [30], which affects physical and cognitive performance [59, 60]. In this study, the fixed-base driving simulator is paired with an AR HMD to conduct the experiments, which may further increase the severity of motion sickness [29]. Out of the 21 participants, 4 participants experienced severe motion sickness for which the experiment was prematurely terminated. The rest of the participants mentioned to have experienced no motion sickness at all, or mild symptoms of motion sickness, which was mutually consented for continuation. However, in a study by Vovk et al. (2018), only negligible symptoms of simulator sickness have been found using the Microsoft HoloLens [68]. Moreover, there is no evidence in contemporary research for an enhanced likelihood of motion sickness in the synergistic operation of an AR HMD and a fixed-base driving simulator. Therefore, the validity of the data affected by motion sickness was treated in the same circumstances as strictly a fixed-base driving simulator, although more research is needed to fully account for the synergistic operation and effects on motion sickness.

## 6 Conclusions

In this research, a concept for an AR driver-assistance system was developed. This system aims to enhance driver safety by providing real-time AR projections related to collision risk.

Two AR design prototypes of the concept were modelled using Unity game engine and subsequently deployed onto a HMD for visual projection. Additionally, a fixed-base driving simulator has been developed using the Carla platform. This enabled real-time data communication between the systems, facilitating the evaluation of the impact of AR projections in a user study involving 17 participants.

The concept of projecting risk areas may offer potential regarding safer driving behaviour by enhancing the awareness to potentially dangerous situations. Based on the results from the analysis of quantitative data extracted from the driving simulator, and qualitative data from the participant interviews, it has been found that the AR projections of risk areas during driving lead to a consistently safer distance keeping, as indicated by the increase in mean time headway at the event of abrupt braking by the lead vehicle. From the data of the questionnaire and interviews, it is concluded that AR projections from Design 1 enhance the attractiveness of a driving system in comparison to no AR visualizations at all. Moreover, there was a strong preference for Design 1 over Design 2, for which the interview remarks suggest that it may be linked to higher functionality, as the projection could redirect the user's attention to the risk of collision. The difference in behaviour may have also led to higher intuitiveness, as the dimension of the projected risk area is consistent with the plane of vehicular movement. However, further research that objectively tests these variables is required to validate this claim.

During the abrupt braking scenarios, no significant differences were found in brake response time, maximum deceleration, and minimum TTC between the conditions. This suggests that the AR projections used in this study did not lead to an improvement in road safety during such critical events. One possible reason for this lack of significance in the data could be that the change in the size of the risk area at the time of a critical event may not have been salient enough to elicit a significant response from the participants. Additionally, the experimental design, which exposed participants to a higher frequency of critical events than they would encounter in real-life driving, might have made participants more cautious and less responsive to the AR projections.

One of the hypotheses was that AR projections lead to a higher amount of necessary yielding when approaching unsignalized intersection scenarios when compared to no AR projections at all. The analysis revealed that only Design 2 increased the yielding success compared to the baseline. However, it's important to note that this result

cannot be regarded as definitive, especially since it was not observed for Design 1. The observed effect may be a result of order effects due to the lack of randomization or counterbalance in the study design.

Concerning the assumption of an increase in cognitive load, there was no difference in the subjective response between the conditions regarding the driver's mental workload. Thus, the extent of AR risk areas as shown in this study affecting the cognitive load is negligible. However, the subjective data did reveal an increase in interference regarding the driving task when AR projections of Design 2 was projected as compared to Design 1 and the baseline.

These findings suggest that future research in AR risk visualization should explore various design elements, including the formula used to determine the dynamics, the location, color, and transparency of the projections. These design considerations are crucial to minimize potential interference and address safety implications related to redirecting the driver's attention.

Finally, it is advisable to explore alternative research approaches, including the comparison of AR risk visualization with auditory or tactile warning systems, to assess their effectiveness and user preferences in different driving scenarios. To enhance the experiment design, future studies should consider reducing the frequency of critical events in a short period of time to improve the reliability and credibility of the simulation data. Coupling this approach with a larger sample size and randomized participant assignment to mitigate order effects could provide stronger evidence regarding the enhancement of safer driving behavior and road safety through AR risk visualization.

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## A Time Table of the User Study

| Task                                       | Description                                       | Duration    |
|--------------------------------------------|---------------------------------------------------|-------------|
| Study preparations<br>Briefing (Session 1) | Informed Consent, Privacy statement, Demographics | 5 minutes   |
| Adjusting HoloLens and Simulation seat     | Find comfortable fit                              | 1 minutes   |
| Familiarization session                    | Simulator/interaction test drive                  | 3 minutes   |
| Break & Questions (Optional)               |                                                   | (5) minutes |
| Drive (A)                                  | Baseline                                          | 5 minutes   |
| Questionnaires                             | UEQ / DALI                                        | 2 minutes   |
| <b>Briefing (Session 2)</b>                | Design 1                                          | 1 minutes   |
| Familiarization session                    | Simulator/interaction test drive                  | 3 minutes   |
| Break & Questions (Optional)               |                                                   | (5) minutes |
| Drive (B)                                  | Intervention                                      | 5 minutes   |
| Questionnaires                             | UEQ / DALI                                        | 2 minutes   |
| <b>Briefing (Session 3)</b>                | Design 1                                          | 1 minutes   |
| Drive (A)                                  | Intervention                                      | 5 minutes   |
| Questionnaires                             | UEQ / DALI                                        | 2 minutes   |
| Break & Questions (Optional)               |                                                   | (5) minutes |
| <b>Briefing (Session 4)</b>                | Design 2                                          | 1 minutes   |
| Drive (B)                                  | Intervention                                      | 5 minutes   |
| Questionnaires                             | UEQ / DALI                                        | 2 minutes   |
| Break & Questions (Optional)               |                                                   | (5) minutes |
| <b>Briefing (Session 5)</b>                | Design 2                                          | 1 minutes   |
| Drive (A)                                  | Intervention                                      | 5 minutes   |
| Questionnaires                             | UEQ / DALI                                        | 2 minutes   |
| Break & Questions (Optional)               |                                                   | (5) minutes |
| <b>Briefing (Session 6)</b>                |                                                   | 1 minutes   |
| Drive (B)                                  | Baseline                                          | 5 minutes   |
| Questionnaires                             | UEQ / DALI                                        | 2 minutes   |
| Interview                                  | Open questions for extra insights                 | 5 minutes   |
| Appreciation                               |                                                   | 1 minute    |
| Total                                      |                                                   | 65 + (25)   |